GazeProtector: On-Device Privacy-Preserving Gaze Prediction for Real-Time Extended Reality

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Abstract

Modern extended reality (XR) headsets integrate multimodal sensors, such as eye- and motion-tracking, to capture fine-grained behavioral cues for immersive interaction. Among these, eye-tracking is especially important for core utility tasks such as gaze prediction. However, the sensitivity of eye-tracking data raises serious privacy concerns, as adversaries can exploit raw eye images or high-fidelity gaze streams to infer personal information and perform successful re-identification attacks (RDA). These risks become more severe when such data are offloaded to *cloud-hosted* AI services for inference, where they may be intercepted or misused by untrusted third parties. Existing defenses, such as differential privacy (DP), inject task-agnostic noise; however, applying uniform noise to high-dimensional XR data can unnecessarily perturb task-relevant features, reduce utility, and increase latency, limiting real-time XR deployment. Motivated by this, we propose a novel privacy mechanism, *GazeProtector*, which is based on an on-device encoding framework that learns to separate and transmit only utility-relevant features for gaze prediction while suppressing identity and other sensitive attributes. The learned encoder operates *before* any storage or transmission, preventing direct visual inspection or accurate reconstruction of the original eye signal and releasing only compact task-specific features to downstream models. This reduces the attack surface (e.g., re-identification) without relying on external noise injection. We evaluate *GazeProtector* on three state-of-the-art deep learning (DL)-based gaze-prediction models and two public XR gaze datasets, DGaze and OpenEDS2020. Our results show strong utility-privacy trade-offs: the proposed mechanism achieves mean angular error (MAE) of 3.71° and 3.81° on DGaze and OpenEDS2020 datasets while reducing re-identification rates by 63.6% and 64.4% and improving end-to-end inference latency by up to ≈2× compared to a traditional DP-based privacy mechanism. We further deploy the mechanism on an HTC Vive Pro headset, demonstrating real-time privacy-preserving gaze prediction during gameplay. These results establish on-device learning as an effective and practical alternative to noise-based defenses for dependable real-time XR systems.

CCS Concepts

• **Human-centered computing → HCI theory, concepts and models**; **HCI design and evaluation methods**.

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Keywords

On-Device Privacy Preservation, Deep Learning, Class Activation Maps, Extended Reality, Re-identification Attack, Gaze Prediction

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1 Introduction

Modern extended reality (XR) headsets are equipped with a suite of multimodal sensors (e.g., eye-tracking, motion-tracking, etc.,) that capture rich behavioral and physiological cues to enable immersive and adaptive user experiences. Among these, eye-tracking plays a particularly crucial role, as it records precise gaze directions and movement patterns that reflect a user's cognitive processes, visual attention, and perceptual state [3, 8, 20]. Leveraging these data, state-of-the-art (SOTA) artificial intelligence (AI) techniques, particularly machine learning (ML) and deep learning (DL) algorithms, are widely used in XR for various utility tasks, such as real-time gaze prediction [25], foveated rendering [38], iris recognition [29], gaze-based identification [61], etc., Prior studies show that users can be identified from only a few seconds of gaze data with high accuracy and low equal error rates [8, 11, 61]. Beyond gaze coordinates, such data can implicitly encode sensitive attributes, including identity, gender, cognitive and emotional states, health conditions, and sexual orientation [8, 11, 37, 64]. This surplus information can be exploited by untrusted service providers or malicious adversaries, leading to unintended information leakage or identity exposure. Furthermore, this introduces the risk of re-identification in captured data through re-identification attacks (RDA), in which XR users' identities are revealed by linking anonymized data to their real-world identities using auxiliary information (e.g., behavioral patterns) captured alongside the data [11, 37]. Unlike traditional identifiers (e.g., usernames), gaze and other behavioral signals are generated subconsciously and are difficult for users to control or suppress. This lack of control highlights the urgent need for privacy-preserving mechanisms to protect eye-tracking data and safeguard user privacy in XR environments.

Researchers have applied various privacy mechanisms, such as differential privacy (DP), k-anonymity, kalε-ido, Gaussian noise, local DP (LDP), and composite approaches [11, 12, 26] for different gaze-data representations, including time-series gaze positions [34] and eye images [59]. Although these methods provide theoretical privacy guarantees, they have practical limitations for XR. Since DP introduces indiscriminate noise, which is particularly problematic in XR, where eye-tracking data are already noisy sources of information, adding more noise to preserve privacy may ruin the user experience. Furthermore, these methods often introduce hardware overhead and computational delays, limiting their suitability for real-time XR tasks, while their application-specific designs hinder generalization across diverse XR use cases. In addition, many existing defenses are application-specific and do not generalize well across diverse XR use cases. Very recently, Du et al. [14] proposed a privacy-preserving approach that transforms full-face RGB images into obfuscated versions while preserving gaze-estimation performance. However, this method is designed for full-face inputs and applies obfuscation only after raw images are captured, making it incompatible with modern XR eye-tracking systems, which use near-eye IR imagery rather than facial images and thus pose fundamentally different privacy risks. Since sensitive information remains exposed during acquisition, post-capture sanitization creates a window of vulnerability before protection is applied. Thus, this approach is unsuitable for SOTA XR headsets, where privacy must be enforced directly at the device level rather than through downstream image obfuscation. These limitations motivate the need for a fundamentally different privacy strategy for XR, where utility-relevant features, such as gaze direction, are separated from privacy-sensitive cues, such as identity, at the point of capture. By enforcing privacy directly on-device, sensitive data are never digitally exposed, transmitted to cloud-based AI models, or stored for later use. This reduces attack surfaces for re-identification and sensitive attribute inference while preserving low latency, bandwidth efficiency, and high model accuracy for utility tasks (e.g., gaze prediction), all of which are essential for real-time XR experiences.

Contribution. This paper presents a novel privacy mechanism, *GazeProtector*, which is based on an on-device encoding framework that learns to separate and transmit only utility-relevant features for gaze prediction while suppressing identity and other sensitive attributes. The primary goal is to develop an edge-based encoder that preserves utility-relevant features while suppressing private information in XR simulation data. The framework consists of two key stages: *feature separation* and *masking*. In the feature separation step, Class Activation Maps (CAMs) identify task-specific regions within the input XR data, enabling the encoder to retain the information most relevant to gaze prediction. In the masking step, privacy-sensitive features are selectively blocked so that only utility-relevant information is transmitted. This CAM-guided masking suppresses private information during data acquisition while preserving gaze-prediction utility, improving computational efficiency, reducing inference time, and strengthening privacy protection. Unlike traditional DP-based privacy methods [11, 54], which inject noise into features and samples during real-time inference and often degrade performance while increasing computational overhead, *GazeProtector* minimizes noise injection, improves efficiency, and maintains high task utility. Specifically, we first develop three DL models, i.e., Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Transformer, for real-time gaze prediction in XR environments. We then introduce a privacy-preserving encoder with an anchor loss function to encourage task-specific feature separation in the encoded representation. Using CAMs, the encoder identifies gaze-relevant features and selectively masks privacy-sensitive cues during XR data readout, thereby protecting user privacy while preserving the essential information needed for accurate gaze prediction. We evaluate the proposed privacy mechanism on two SOTA public XR gaze datasets: DGaze [25] and OpenEDS2020 [56]. The results show that integrating *GazeProtector* with DL-based gaze prediction preserves high utility while protecting XR user privacy. For instance, with CAM-guided 25% masking, the Transformer model achieves mean angular errors (MAE) of 3.71° and 3.81° on the DGaze and OpenEDS2020 datasets, which are almost identical to the results obtained without any privacy mechanism. We further evaluate the proposed method against a well-known RDA. Under RDA, the same setting reduces re-identification rates by 63.6% and 64.4% on DGaze and OpenEDS2020 datasets, respectively. Finally, we deploy *GazeProtector* on a consumer-grade XR headset, HTC VIVE Pro, for real-time gaze prediction in foveated rendering tasks during XR gameplay. The deployment results show that the proposed mechanism reduces inference time by up to $\approx 2\times$ compared with traditional DP-based privacy mechanisms while maintaining high gaze-prediction utility and protecting user privacy.

2 Related Works

Eye Tracking and ML/DL-Based Gaze Prediction in XR: Modern XR systems combine visual, haptic, auditory, and inertial sensors to capture rich behavioral and environmental information, with eye tracking emerging as a vital modality. Eye movements capture users' physical, cognitive, and emotional states and enable natural interaction (e.g., object selection and cursor control) via gaze-based pointing and fixations [8, 50, 58]. Gaze is also fundamental to foveated rendering [25, 38, 40], which prioritizes high-resolution rendering in the fovea to reduce computation while preserving visual quality that is critical for resource-constrained XR systems. Beyond rendering, synchronized eye–head movements enhance avatar realism and social presence [8], and gaze analytics supports behavioral studies such as AOI detection, task analysis, and attention visualization [8, 10, 22]. In parallel, ML and DL have transformed gaze prediction, outperforming traditional appearance-based methods by leveraging multimodal patterns from large XR datasets (e.g., eye images, head pose, saliency features, etc.,) [8, 9, 24, 25, 27, 38, 43, 44]. For instance, Hu et al. [25] developed a CNN-based model integrating object position sequences, head velocity, and saliency cues for foveated rendering; Melnyk et al. [43, 44] employed LSTM and Transformer architectures for gaze prediction; and Liu et al. [38] adopted Vision Transformer for similar tasks. While these DL models advance and improve interaction, rendering efficiency, and behavioral understanding, gaze signals also encode sensitive biometric and behavioral cues, raising substantial privacy and security risks.

Privacy Risks in XR: Privacy is a critical concern in XR applications because these systems collect sensitive, multimodal data (e.g., eye-tracking). Although these data are essential for interactivity and rendering efficiency, gaze data can be exploited for profiling or targeted advertising [26]. Research has shown that gaze patterns can reveal personal attributes such as age [37, 77], gender [11, 60], racial background [6, 26], and even neurological or behavioral conditions [8, 66], making gaze data inherently identifiable. Recent studies also demonstrate domain-specific re-identification across XR tasks [48, 51, 52, 55]; for instance, David et al. [10, 12] identified users across 360° scenes using fixation and saccade features, and Nair et al. [54] showed large-scale identification in VR games (e.g., *Beat Saber*). Beyond gaze, other behavioral signals pose risks. For instance, Liu et al. [37] found re-identification vulnerabilities in facial blendshapes, Meng et al. [46] showed VRChat motion patterns can de-anonymize users, while Yang et al. [76] inferred typed content by observing avatar movements. These privacy implications underscore the urgent need for secure handling of gaze data, particularly in XR applications, where the potential for re-identification presents a substantial risk to users' privacy.

Privacy-Preserving Mechanism in XR: Many privacy mechanisms have been developed to address privacy risks in XR. Recently, formal methods such as DP [7, 11, 12, 34, 36] have attracted the attention of XR communities for their ability to protect released data (e.g., eye-tracking data) from identification or sensitive inference by constraining output distributions. Other privacy mechanisms, such as Gaussian and Exponential mechanisms, have been applied to saliency heatmaps [36] and aggregated gaze features [65]. In contrast, other approaches, such as k-anonymity, kalϵ-ido, Gaussian noise, and composite methods [11, 26] have been employed to safeguard time-series gaze data [34] and eye images [59]. While these methods offer theoretical privacy guarantees, they face practical challenges, as data correlations can compromise privacy. For instance, David et al. [11, 12] applied DP, k-anonymity, and γ-plausible deniability to mitigate re-identification risks in gaze prediction tasks. Although these approaches effectively reduce privacy leakage with minimal training error, their performance is highly application-dependent. Moreover, it should also be noted that eye-tracking data is inherently noisy; adding additional noise to preserve privacy can degrade the user experience. Beyond formal mechanisms, reinforcement learning (RL) has been explored to reduce sensitive inferences during utility optimization [18], but RL methods remain probabilistic and lack formal guarantees. Federated learning (FL) [15, 17, 75] improves privacy by keeping data local, yet provides no cryptographic protection comparable to DP or secure aggregation, and struggles with scalability on high-dimensional XR data. Du et al. [14] recently proposed a full-face RGB obfuscation method for privacy-preserving gaze estimation. However, because it sanitizes data only post-capture and XR systems rely on near-eye IR imagery, the approach exposes sensitive information at acquisition and is incompatible with modern XR pipelines. Existing defenses are also highly application-specific and introduce substantial computation overhead, making them unsuitable for real-time XR gaze prediction. These limitations highlight the need for a sensor-level strategy that enforces privacy at the point of capture, suppressing identity-related cues before they enter the digital domain, which motivates our work.

3 Evaluating the Privacy Leakage of DL-based Gaze Prediction Application

In this section, we present the details of our threat model. We focus on privacy leakage in AI-driven XR applications by analyzing re-identification attacks (RDA) against developed XR DL models for gaze prediction tasks. We provide brief information about the threat model for RDA against XR DL models as follows.

Re-identification attack: An RDA occurs when an adversary successfully matches anonymized data to a specific individual by leveraging auxiliary information such as demographics or behavioral features. Even if personal identifiers (i.e., names and birthdates) are removed from a dataset, remaining attributes such as age, gender, and behavioral data (e.g., saccades, fixations, etc.,) can reveal unique patterns that enable re-identification. For instance, if a dataset retains eye-tracking data and demographics, an adversary can select the identities that match the demographics of their target and then train and apply a model to re-identify individuals.

Threat Model: The threat models we consider for RDA in this paper are inspired by [10, 11, 37]. We consider an XR system in which XR sensors, such as eye-tracking integrated into head-mounted displays (e.g., HTC VIVE PRO, Meta Quest Pro, Magic Leap 2, etc.,), interact with a third-party service provider (e.g., a cloud host) to perform inference tasks (e.g., gaze prediction). In this setup, raw sensor eye-tracking data representing gaze information is captured by the XR device and transmitted to the host, which processes the data using a DL model to perform a gaze-prediction task. For the RDA, we assume that an adversary has access to a public XR dataset containing gaze-related features, such as saccade amplitudes, fixation durations, and gaze vectors, along with auxiliary demographic attributes (e.g., age or gender). The attacker's goal is to infer the identities of anonymized users solely from these behavioral signals. To achieve this, the adversary trains an identification model to take the XR dataset gaze-related features as input and output the associated identity. Given a new sample from the dataset, the attacker applies the trained model to infer the XR user identity. The RDA attack is successful if the model correctly identifies the individual. To execute the RDA on DL-based gaze prediction tasks, we make a similar assumption proposed by David et.al. [10, 11]. This formulation assumes that the attacker does not require direct access to explicit personal identifiers (such as names or unique IDs). Instead, the attack leverages residual correlations in gaze data and associated behavioral signals that persist after anonymization. The central insight driving this threat model is that eye-tracking patterns are highly individualistic, functioning as behavioral biometrics, thus providing sufficient information for re-identification even in the absence of direct identifiers. Therefore, any XR gaze model or dataset that retains such identity-linked information remains susceptible to re-identification risks. This threat model underscores the need to evaluate and mitigate identity leakage in XR systems to ensure the privacy-preserving deployment of AI-driven gaze prediction models.

4 Proposed GazeProtector Methodology

In XR, gaze prediction supports real-time estimation, foveated rendering, and adaptive interaction, but eye-tracking images and derived trajectories can also encode sensitive attributes such as identity, demographics, and cognitive states. To balance utility and privacy, we design a framework that preserves gaze-prediction performance while minimizing leakage on private tasks (e.g., identity or gender classification). As shown in Figure 1, the proposed method has three phases: (1) Developing DL-enabled gaze prediction models in XR, (2) a CAM-guided encoder for mask generation, and (2) A two-stage adversarial training regime. The details are described below.

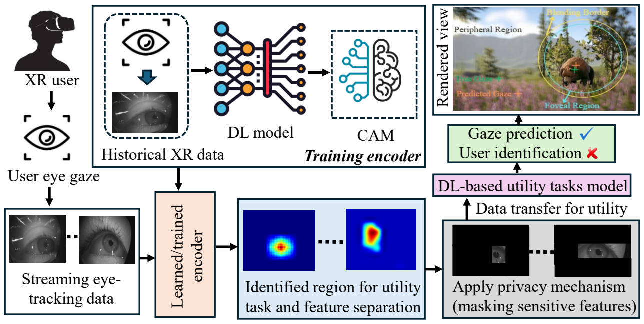

Figure 1: Overview of the proposed GazeProtector privacy mechanism for real-time gaze prediction in foveated rendering systems, which encodes sensitive eye-tracking data directly within the XR device to prevent raw gaze information from being exposed during inference, thereby ensuring both privacy protection and real-time performance.

4.1 Problem Setup

Let the raw eye-tracking data be represented by a tensor $X \in \mathbb{R}^{N \times D}$, where N is the number of spatial or temporal features (e.g., pixels or time-steps) and D is the feature dimension. For each input X, there is a corresponding utility label y_u, representing the target gaze coordinate. A private attribute y_p (e.g., user identity) can be inferred from X, but is not explicitly available as a label to our system during utility training. Our framework aims to learn a data-driven, task-specific *masking function*, G_ϕ, which takes an input X and produces a soft mask $M = G_\phi(X)$, where $M \in [0, 1]^N$. This mask is applied element-wise to the input to produce a sparsified version, $X' = M \odot X$. A gaze predictor, U_{θ_U}, is then trained to predict the target gaze coordinate from this masked input: $\hat{y}_u = U_{\theta_U}(X')$. The core of our approach is governed by a user-defined *privacy-utility budget*, $k \in [0, 1]$, which dictates the maximum allowable fraction of features that can be preserved. A smaller k corresponds to more aggressive masking and thus higher privacy. Our objective is to find the optimal parameters for the mask generator, ϕ^*, and the gaze predictor, θ_U^*, that minimize the gaze prediction error while ensuring that an adversary, A_{θ_A}, cannot succeed. We formalize this as a constrained adversarial optimization problem over a utility dataset $\mathcal{D}_u$ of (X, y_u) pairs:

$$\phi^*, \theta_U^* = \arg\min_{\phi, \theta_U} \left(\mathbb{E}_{(X,y_u)\sim\mathcal{D}_u} [\mathcal{L}_{gaze}(U_{\theta_U}(G_\phi(X) \odot X), y_u)] \right), \quad (1)$$

subject to two key constraints:

$$||G_\phi(X)||_1 \le k \cdot N, \quad \forall X \in \mathcal{D}_u \quad \text{(User-defined Privacy Budget)}$$

$$\max_{\theta_A} \mathbb{E}_{X\sim\mathcal{D}_u} [\text{Accuracy}(A_{\theta_A}(G_\phi(X) \odot X))] \le \mathcal{A}_{chance} \quad \text{(Adversarial Failure)}$$

Here, $\mathcal{L}_{gaze}$ is the loss for the gaze prediction task, typically the Mean Squared Error (MSE) or Mean Angular Error (MAE) between the predicted and true gaze coordinates. The adversarial failure constraint in Eq. Adversarial Failure ensures that the best possible adversary, trained on the masked data, cannot achieve an accuracy significantly above the chance level, $\mathcal{A}_{chance}$. This formulation forces the mask generator G_ϕ to learn which eye-movement dynamics are essential for predicting future gaze trajectories while being uncorrelated with biometric identifiers, all while adhering to the user's explicit privacy budget k.

4.2 DL-enabled Gaze Prediction Models in XR

The foremost step in our proposed methodology is to develop DL models for gaze prediction as a utility task. We train three DL models for this purpose: CNN, LSTM, and Transformer. These models are chosen based on their widespread adoption and demonstrated efficacy in SOTA DL-enabled XR gaze prediction research [11, 25, 38, 43, 44]. The utility predictor U_{θ_U} and the adversary A_{θ_A} can be implemented using any of these architectures. The CNN comprises two convolutional layers followed by max-pooling and two dense layers. The LSTM uses six stacked 128-unit LSTM layers with recurrent dropout, followed by two dense layers with ReLU activation. The Transformer follows a standard sequence-to-sequence architecture [70] that consists of three components: (i) a positional encoding layer to embed input sequences and encode temporal information, (ii) an encoder with multi-head self-attention to capture key patterns in XR data, and (iii) the output layer that predicts gaze through an activation, using a 64-dimensional attention representation with normalization and a 256-unit feed-forward MLP to predict final gaze coordinates. The mask generator G_ϕ is implemented with a lightweight architecture (e.g., a shallow CNN or MLP) suitable for on-device execution. For the gaze prediction utility task, the loss function $\mathcal{L}_{utility}$ is MSE.

4.3 CAM-Guided Mask Generation

To effectively train the mask generator G_ϕ, we require a guidance signal to teach it which features are important for the utility task versus the private task. We derive this signal using task-specific saliency maps generated via Gradient-weighted Class Activation Maps (Grad-CAM) [62]. For each feature position $u \in \{1, \ldots, N\}$, let $h_u \in \mathbb{R}^d$ be the latent feature vector from a pre-trained baseline model (where d is the latent dimension). For a target task $c \in \{T_1, T_2\}$ with pre-softmax score y^c, the importance weight α_u^c for position u is given by $\alpha_u^c = \frac{1}{d}\sum_{k=1}^{d} \frac{\partial y^c}{\partial h_{u,k}}$. The corresponding saliency map $L^c(u) = \text{ReLU}(\sum_{k=1}^{d} \alpha_u^c \cdot h_{u,k})$ is normalized to create a stable guidance signal $\tilde{L}^c$. We then define two regularization losses for training the mask generator:

- A *CAM-alignment loss*, $\mathcal{L}_{\text{CAM}} = -\sum_u M_u \tilde{L}^{T_1}(u) + \beta \sum_u M_u \tilde{L}^{T_2}(u)$, encourages the mask M to align with the utility map $\tilde{L}^{T_1}$ while misaligning with the private map $\tilde{L}^{T_2}$. The hyperparameter $\beta > 0$ controls the strength of private feature suppression.
- A *budget constraint loss*, $\mathcal{L}_{\text{budget}} = \max(0, \frac{1}{N}\|M\|_1 - k)$, provides a soft penalty to enforce the user's budget.

4.4 Two-Stage Adversarial Training Regime

We adopt a two-stage training procedure to stably learn the mask generator and utility predictor.

Stage 1: Baseline Model Training for Guidance. First, we train high-performance baseline models for the utility task (U_{θ_U}) and a privacy task (A_{θ_A}) using unmasked data to generate static guidance maps. A significant challenge is the absence of explicit private labels (e.g., demographics) in most public datasets. To overcome this, we adopt the participant's subject ID as a powerful proxy for their unique biometric signature. Research has consistently shown that an individual's gaze dynamics—such as saccadic paths, fixation durations, and gaze fingerprint that is highly distinctive and can be used for reliable user identification [11, 53]. Consequently, a privacy adversary A_{θ_A} trained to classify users based on their subject ID is not merely memorizing random labels; it is learning to recognize these idiosyncratic, biometric patterns. The resulting Grad-CAM guidance map, $\tilde{L}^{T_2}$, therefore, isolates the specific temporal or spatial features that are most indicative of this biometric signature. This stage is performed only once on a public dataset with available subject IDs, providing strong guidance for the main training process without requiring any private labels for the target users' data.

Stage 2: Joint Adversarial Learning of Mask and Predictor. In this stage, we optimize the mask generator G_ϕ and the utility predictor U_{θ_U} on the user's gaze data by solving a practical, unconstrained version of our problem statement. The full objective combines the utility loss, the CAM and budget regularization terms, and an adversarial term in a minimax objective:

$$\begin{aligned}\min_{\phi,\,\theta_U}\max_{\theta_A}\;& \mathcal{L}_{utility}(U_{\theta_U}(M\odot X), y_u) + \lambda_{\text{CAM}}\mathcal{L}_{\text{CAM}} \\ &+ \lambda_{\text{bud}}\mathcal{L}_{\text{budget}} - \gamma\mathcal{L}_{privacy}(A_{\theta_A}(M\odot X), y'_p).\end{aligned} \tag{2}$$

Here, $\mathcal{L}_{privacy}$ is the adversary's loss (e.g., binary cross-entropy), and $\lambda_{\text{CAM}}, \lambda_{\text{bud}}, \gamma$ are hyperparameters balancing the different objectives. The inner maximization trains a strong adversary on the masked data $M \odot X$ (where y'_p are proxy labels if real ones are unavailable). The outer minimization updates G_ϕ and U_{θ_U} to maximize utility while rendering the masked data useless to the adversary. This minimax game is typically solved by alternating gradient steps on θ_A and (ϕ, θ_U).

4.5 Practical Implementation and Benefits XR

Once trained, the lightweight mask generator G_ϕ and the utility predictor U_{θ_U} are deployed. In a real-time XR setting, the raw XR sensor data (e.g., eye-tracking) X is first passed through G_ϕ to produce the mask M according to the user-set budget k. The masked data $X' = M \odot X$ is then processed by U_{θ_U} for the utility task. This approach yields two significant practical advantages. Primarily, it offers a *tunable privacy-utility trade-off*, as the user or system can dynamically adjust the privacy level in real-time by simply changing the budget parameter k. A lower k leads to a sparser input (higher privacy, potentially lower utility), while a higher k preserves more data. Furthermore, this data sparsification inherently provides enhanced *computational efficiency*. By reducing the number of active features, our method lowers the computational load for downstream models—fewer time steps or fewer active pixels for an LSTM, Transformer, or CNN to process—which in turn reduces both data transmission bandwidth and the number of operations required for inference.

5 Datasets & Experimental Setup

This section provides an overview of the evaluation metrics, experimental setup, and dataset we used in our proposed approach.

5.1 Dataset

To evaluate the effectiveness of our proposed privacy mechanisms for gaze prediction, we use two open-source XR gaze datasets: OpenEDS2020 [56] and DGaze [25]. Note that these datasets are widely used in XR security and privacy research and have served as standard benchmarks for gaze prediction tasks [4, 11–13, 38, 41, 71, 75]. The details of these datasets are summarized below.

OpenEDS2020: The OpenEDS2020 [56] dataset comprises eye-tracking data from 80 participants (56 male, 24 female), ages 20–70 years. It contains 128,000 images for training and 70,400 images across the validation and test sets. All participants wore a VR head-mounted display (HMD), and images were captured by a near-eye camera operating at 100 Hz with a resolution of 640×400 pixels. All images are 8-bit grayscale (i.e., each pixel is a single value in $[0, 255]$). The dataset includes ground-truth 3D gaze vectors, which we converted to 2D (horizontal and vertical components), following prior work to facilitate evaluation [38].

DGaze: The DGaze [25] dataset comprises VR eye-tracking recordings sampled at 100 Hz from 43 participants (25 male, 18 female; ages 18–32) who freely explored dynamic 3D virtual scenes. Each timestamp provides the gaze position along with synchronized scene annotations and signals, including saliency features, tracked-object metadata (identity, position, and motion cues), head kinematics, and aligned scene screenshots. The dataset contains 86 user–scene recordings ($\approx$ 180 s each), yielding per recording approximately 18,000 gaze samples at 100 Hz, 18,000 object-position samples at 100 Hz, 36,000 head-velocity samples at 200 Hz, and 10,800 screenshots at 60 fps, with all streams timestamped for precise cross-modal alignment. These synchronized modalities capture both oculomotor behavior and scene dynamics, making DGaze well-suited for modeling short-term gaze behavior in XR environments

Data Preprocessing: For the *OpenEDS2020* dataset, we first horizontally flipped the right-eye images to align them with the left-eye images, enabling a unified gaze-mapping function for both eyes [38]. A cropping parameter of $\beta_1 = 35$ was applied to isolate the eye region and remove redundant background pixels. Finally, all images were normalized and standardized to ensure consistent input distributions for the DL-based gaze prediction model. For the *DGaze* dataset, we first resampled all data streams (gaze vectors, head kinematics, and object metadata) to a common 100 Hz temporal frequency to ensure precise alignment with the gaze signal. Gaze vectors were normalized within [0,1] relative to the scene resolution, and all features were standardized before concatenation into the final model input representation.

5.2 Experimental Setup

We used PyTorch 2.3 [57] for training and evaluating our DL-based gaze prediction models. For Grad-CAM application, we utilized the Advanced AI explainability library for PyTorch [21]. All models were trained on a system equipped with an Intel Core i9 Processor, 128GB RAM, and an NVIDIA GeForce RTX 3090 Ti GPU.

Hyperparameters: We trained all DL models using the Adam optimizer for up to 250 epochs with a learning rate of 3×10^{-4}, using batch sizes of 512 (OpenEDS2020) and 256 (DGaze), and applied early stopping with a patience of 30 to mitigate overfitting. We use a Radial Basis Function Network (RBFN) to evaluate identification rates under RDA for these DL-based gaze prediction models, similar to the work [10, 11]. The RBFN, with a single hidden layer of activation functions, generates class probabilities based on input features and identifies the class with the highest probability as the

Table 1: Baseline performance of gaze-prediction models without privacy mechanisms, measured by MAE in degrees. The X/Y represents the DGaze/OpenEDS2020 datasets.

Models	100ms	200ms	300ms
LSTM	4.85/5.06	6.03/5.76	7.58/6.79
CNN	4.13/3.98	5.82/5.43	7.53/6.39
Transformer	3.45/3.68	4.44/4.47	6.65/5.34

most likely match for biometric identification. Feature vectors from an unknown individual are classified across multiple stimuli, with prediction scores averaged to determine identity. The identification rate is reported as the percentage of correctly classified individuals, averaged over 40 runs with fixed splits: 50/50 for the DGaze dataset and 70/30 for the OpenEDS2020 dataset.

Evaluation Metrics: We use the most widely used performance metrics to evaluate the performance of gaze prediction models using mean angular error (MAE) between the ground truth and the predicted gaze position. The smaller the MAE value, the better the gaze prediction performance. To assess the effectiveness of our proposed RDA privacy attack, we report the identification rate, similar to the work [10, 11]. Additionally, we compare the privacy-utility trade-off using the MAE, which measures the impact of our privacy mechanism on gaze prediction performance under RDA conditions.

6 Experimental Results

This section presents the results of our proposed GazeProtector privacy method for DL-enabled gaze-prediction utility tasks and its effectiveness against RDA attacks in protecting XR users' privacy.

6.1 Gaze Prediction without Privacy Mechanism

We evaluate the gaze prediction performance regarding the MAE metric of different prediction models (e.g., CNN, LSTM, and Transformer) over different prediction windows (e.g., 100ms, 200ms, and 300ms) using the DGaze and OpenEDS2020 datasets, as shown in Table 1. We begin by presenting the performance of the prediction models without applying the proposed privacy mechanisms. From Table 1, we observe that the Transformer model has a smaller MAE value than the CNN and LSTM models for different prediction windows for both datasets. For instance, using the DGaze dataset, the Transformer model achieves an MAE value of 3.45° when the window length is 100 ms, which is 1.41× and 1.12× lower than the LSTM and CNN models using the same prediction window. A similar phenomenon is observed for the OpenEDS2020 dataset across all prediction models and different prediction windows.

6.2 Grad-CAM Guided Feature Identification and Separation for Gaze Prediction Tasks

We evaluate DL-based gaze-prediction models using Grad-CAM encoders to identify regions that are most relevant to the gaze-prediction task. Figure 2 visualizes Grad-CAM activations from the Transformer model on representative eye images from the OpenEDS2020 dataset. The first row shows the original eye images; the second row shows the Grad-CAM-encoded heatmaps; the third row overlays the heatmaps on the original images; and the fourth row presents the masked regions identified from the CAM-guided masking process. The heatmaps show that the model assigns higher

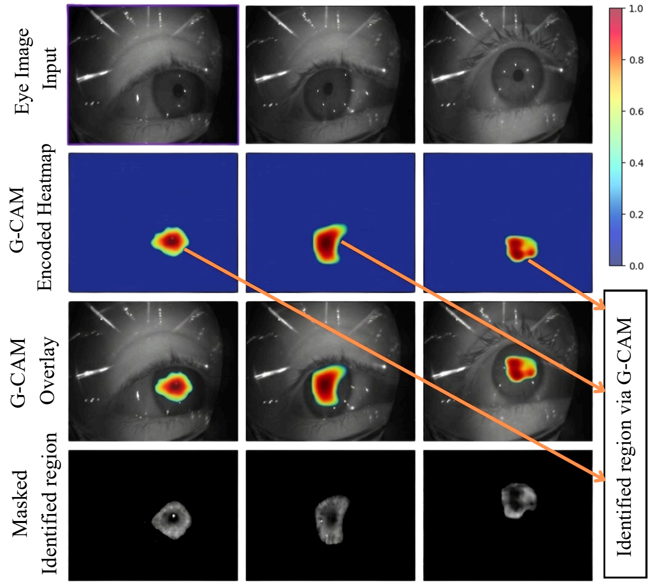

Figure 2: Visualization of Grad (G)-CAM activations from the Transformer model for gaze prediction on three random input eye images from the OpenEDS2020 dataset. The rows (top to bottom) show the original eye images, Grad-CAM heatmaps, Grad-CAM overlays, and masked identified regions. The results demonstrate that the proposed CAM-guided mechanism localizes gaze-relevant regions and selectively masks non-essential or privacy-sensitive areas, enabling privacy-preserving gaze prediction.

Table 2: Gaze prediction models' performance using the GazeProtector privacy mechanism at various masking ratios across three prediction windows on two datasets.

		DGaze (MAE°, ↓)			OpenEDS2020 (MAE°, ↓)		
Mask	**Model**	**100 ms**	**200 ms**	**300 ms**	**100 ms**	**200 ms**	**300 ms**
25%	**LSTM**	4.70	6.21	7.82	4.41	5.88	6.92
25%	**CNN**	4.31	5.92	7.68	4.14	5.51	6.69
25%	**TF**	3.71	4.68	6.89	3.81	4.71	5.72
50%	**LSTM**	4.93	6.48	8.09	4.93	6.13	7.12
50%	**CNN**	4.59	6.11	7.89	4.52	5.95	6.52
50%	**TF**	3.98	5.34	7.04	3.98	4.95	6.06
75%	**LSTM**	5.15	6.88	8.44	5.23	6.37	7.53
75%	**CNN**	4.91	6.55	8.09	4.95	6.14	6.88
75%	**TF**	4.28	5.83	7.45	4.15	5.34	6.42

activation values to gaze-relevant regions, while less informative areas receive low activation. The color scale ranges from 0 (low relevance) to 1 (high task importance). The overlay row further confirms that the Transformer localizes important gaze-related regions within the eye images. Finally, the masked identified regions demonstrate that the proposed mechanism preserves the high-activation task-relevant regions while suppressing non-essential and potentially privacy-sensitive areas. This shows that the proposed objective function effectively guides feature separation, improving interpretability while enabling privacy-preserving gaze prediction. A similar mechanism is applied to the DGaze dataset, ensuring both high gaze prediction accuracy and user privacy.

Table 3: Privacy risk evaluation of the GazeProtector method at a 100 ms prediction horizon across different masking ratios. Each cell reports DGaze/OpenEDS2020 re-identification rates (%, ↓) using RBFN and MLP attack models.

Mask	LSTM		CNN		Transformer	
	RBFN	MLP	RBFN	MLP	RBFN	MLP
None	76.5/72.8	75.0/71.4	73.2/70.5	71.3/70.4	78.4/74.1	76.9/72.5
25%	36.2/33.8	37.4/35.0	32.0/29.9	35.1/34.0	28.5/26.4	31.1/28.5
50%	23.5/21.0	24.4/21.9	21.2/19.3	24.6/22.0	19.8/17.7	22.1/19.9
75%	15.7/14.1	16.4/14.8	14.5/13.4	16.8/15.7	13.2/12.3	15.1/13.9

6.3 Performance Evaluation of GazeProtector

Table 2 summarizes the gaze prediction performance of three DL models (e.g., LSTM, CNN, and Transformer (TF)) using our proposed GazeProtector privacy mechanism across multiple masking ratios (e.g., 25%, 50%, and 75%) and prediction windows (100 ms, 200 ms, 300 ms) on the DGaze and OpenEDS2020 datasets. Overall, the Transformer model consistently outperforms both CNN and LSTM models under all masking conditions. For instance, at a 25% masking ratio, the Transformer achieves the lowest MAE of 3.71°, 4.68°, and 6.89° for 100 ms, 200 ms, and 300 ms prediction windows, respectively, on the DGaze, and 3.81°, 4.71°, and 5.72° on OpenEDS2020 datasets, respectively. We observe that increasing the masking ratio from 25% to 75% gradually degrades performance due to stronger visual suppression. However, the degradation remains modest, indicating that the CAM-guided encoder retains task-relevant gaze features while filtering out identity cues. For instance, at 75% masking, the Transformer model still maintains competitive gaze prediction performance, with MAE values ranging from 4.28° to 7.45° on DGaze and from 4.15° to 6.42° on OpenEDS2020, while outperforming CNN and LSTM models. As expected, MAE also increases with longer prediction horizons due to greater temporal uncertainty. Overall, these results show that *GazeProtector* preserves strong gaze-prediction utility while enabling privacy-aware real-time XR deployment.

6.4 Privacy Evaluation of GazeProtector

Table 3 presents the privacy evaluation of our proposed CAM-guided masking approach at a 100 ms prediction horizon across various masking ratios (None, 25%, 50%, and 75%) for three DL models: Transformer, LSTM, and CNN on DGaze and OpenEDS2020 datasets under RDA. Following prior work [10, 11, 37], we evaluate identification rates for gaze prediction tasks using RBFN and MLP attack models on randomly sampled test data, report the percentage of users correctly identified, and then show the utility of these models for gaze-prediction tasks, both with no privacy and with our proposed privacy-preserving mechanisms, across various masking ratios (e.g., 25%, 50%, and 75%). We observe that, under the no-privacy setting, the Transformer model exhibits the highest re-identification rate across both datasets and RDA models. For instance, in the Transformer model setting, the RBFN model achieves re-identification rates of 78.4% on DGaze and 74.1% on OpenEDS2020, which are approximately 1.03× and 1.02× higher than LSTM, and 1.07× and 1.05× higher than CNN, respectively. Similarly, the MLP model achieves re-identification rates of 76.9% and 72.5% on the same datasets, which are approximately 1.03× and 1.02× higher than LSTM, and 1.08× and 1.03× higher than CNN, respectively. However, we observe that applying our proposed privacy mechanisms significantly reduces re-identification rates across all models and datasets. For instance, with a 25% masking ratio, the Transformer model under the RBFN achieves re-identification rates of 28.5% on DGaze and 26.4% on OpenEDS2020, reducing identification rates by 64% and 66% on both datasets compared to the no-privacy setting. Similarly, under the MLP model, it achieves re-identification rates of 31.1% and 28.5%, corresponding to reductions of approximately 83.2% and 83.4% on DGaze and OpenEDS2020, respectively. Moreover, we observe that increasing the masking ratios further decreases the re-identification rates. For instance, with a 75% masking ratio, the Transformer model reduces identification rates by 83.2% and 83.4% for the RBFN, and by 80.4% and 80.8% for the MLP models, on the same datasets. These results suggest that the proposed privacy mechanism effectively mitigates sophisticated RDA attacks by substantially reducing re-identification rates. A similar phenomenon is also observed for LSTM and CNN models.

6.5 Privacy-Utility Tradeoff Comparison

This section evaluates our proposed GazeProtector privacy mechanisms against RDA, comparing them to the baseline (no privacy) and traditional DP-based privacy methods through a privacy-utility trade-off evaluation, similar to prior work [10, 11, 54]. Note that, since the RBFN-based RDA model shows stronger re-identification capability than the MLP-based attack model, as shown in Table 3, we use RBFN as the primary attack model for the privacy-utility trade-off evaluation. Table 4 reports the MAE value of a 100-ms gaze prediction horizon, along with the identification rate of the re-identification attack for three DL models (e.g., LSTM, CNN, and Transformer) for both the DGaze and OpenEDS2020 datasets across various masking ratios (e.g., 25%, 50%, and 75%) and varying DP privacy levels. For DP, we use three privacy budget (ε) groups (similar to the privacy budget groups in [30, 31, 54]), i.e., high (ε = 1), medium (ε = 3), and low (ε = 5). We observe that the re-identification rate of RDA against the corresponding CAM-guided masking-based privacy mechanisms is not significant, as it is lower than that of both the baseline no-privacy and slightly lower than that of ε-DP-based privacy mechanisms. For instance, using 25% masking ratios, the Transformer model achieves a re-identification rate of 28.5% and 26.4% for the DGaze and OpenEDS2020 datasets, which reduces the identification rate up to 63.6% and 64.4% for the no-privacy mechanisms and 47.5% and 45.0% for traditional DP-based privacy mechanisms with a low ε value (i.e., ε=5). Moreover, we observe that at higher masking ratios (e.g., 75%), our proposed privacy mechanism achieves a lower re-identification rate than traditional DP-based privacy mechanisms at a high ε value (i.e., ε=1), as shown in Table 4. Furthermore, from the utility perspective, our results show that the proposed privacy mechanism has minimal effect on utility. For instance, using masking ratios (25%), the Transformer model achieves an MAE value of 3.71° and 3.81° for the DGaze and OpenEDS2020 datasets, which is almost equal to the no privacy condition and is also lower than traditional DP-based privacy mechanisms with a low ε value (i.e., ε=5), as shown in Table 4. In contrast, it is observed that there is a slight increase in MAE values for higher masking ratios (e.g., 50% and 75%), but the MAE values are lower than a low ε value (i.e., ε=1 and 3). Although GazeProtector

Table 4: Privacy-utility tradeoff comparison across three DL models on the DGaze and OpenEDS2020 datasets at a 100-ms gaze-prediction horizon. Results compare no privacy, CAM-guided masking, and DP with different ε values. Utility denotes MAE° (↓), PRV denotes re-identification accuracy (%, ↓), and X/Y denotes DGaze/OpenEDS2020 datasets.

MD	No Privacy		Mask (25%)		Mask (50%)		Mask (75%)		DP (ε = 5)		DP (ε = 3)		DP (ε = 1)	
	Utility	PRV	Utility	PRV	Utility	PRV	Utility	PRV	Utility	PRV	Utility	PRV	Utility	PRV
LSTM	4.85/5.06	76.5/72.8	4.70/4.41	36.2/33.8	4.93/4.93	23.5/21.0	5.15/5.23	15.7/14.1	5.19/5.45	52.0/48.7	5.51/5.77	39.2/36.0	5.91/6.07	27.1/24.5
CNN	4.13/3.98	73.2/70.5	4.31/4.14	32.0/29.9	4.59/4.52	21.2/19.3	4.91/4.95	14.5/13.4	4.46/4.37	49.8/47.1	4.76/4.55	36.7/34.2	5.08/4.86	25.3/23.0
TF	3.45/3.68	78.4/74.1	3.71/3.81	28.5/26.4	3.98/3.98	19.8/17.7	4.28/4.15	13.2/12.3	3.99/3.96	47.5/45.0	4.18/4.05	34.5/32.4	4.46/4.30	23.4/21.1

Table 5: End-to-end inference latency (ms) at a 100,ms gaze-prediction horizon for three DL models. Results compare no privacy (NP), *GazeProtector* CAM-guided masking (MS), and DP with varying ε values. The X/Y denote DGaze/OpenEDS2020 datasets.

MD	NP	MS 25%	MS 50%	MS 75%	(ε = 5)	(ε = 3)	(ε = 1)
LSTM	3.0 / 3.4	3.1 / 3.5	3.2 / 3.6	3.2 / 3.6	4.4 / 4.8	4.6 / 5.0	5.0 / 5.4
CNN	2.6 / 3.0	2.7 / 3.1	2.8 / 3.2	2.8 / 3.2	3.9 / 4.3	4.1 / 4.6	4.5 / 4.9
TF	3.4 / 3.8	3.5 / 3.9	3.6 / 4.0	3.6 / 4.0	4.7 / 5.1	5.0 / 5.5	5.4 / 5.9

CAM-guided masking results in a slight decrease in utility compared with the no-privacy setting, it substantially lowers privacy risk and offers a favorable privacy-utility tradeoff. Compared with DP, *GazeProtector* more effectively reduces re-identification risk by selectively suppressing identity-bearing regions while preserving gaze-relevant features. Thus, it provides stronger protection against RDA-based privacy leakage while maintaining an acceptable level of gaze-prediction utility for real-time XR applications.

Effectiveness of GazeProtector: Table 5 summarizes the inference time for 100-ms gaze prediction across LSTM, CNN, and Transformer models under three privacy settings: no privacy, CAM-guided masking with 25%, 50%, and 75% masking ratios, and DP-based mechanisms with ε = 1, 3, and 5 on the DGaze and OpenEDS2020 datasets. Overall, the proposed *GazeProtector* mechanism achieves substantially lower inference time than traditional DP-based methods across different ε. For instance, with 75% masking, the Transformer requires only 3.6 ms and 4.0 ms on the CPU for the DGaze and OpenEDS2020 datasets, leading to up to ≈ 2× speed-up in inference compared to the traditional DP-based (ε = 1) method for the same tasks. This shows that CAM-guided masking introduces only marginal overhead, whereas DP increases latency by adding noise directly to gaze samples during inference. Moreover, at lower masking ratios, such as 25% and 50%, inference time remains close to the no-privacy baseline, making *GazeProtector* suitable for real-time, low-latency AI-XR applications. A similar phenomenon is observed for other DL models.

7 GazeProtector Privacy Method Deployment in XR HMD for Real-time Gaze Prediction Task

This section describes the implementation of our proposed GazeProtector privacy mechanism for real-time gaze prediction in the XR simulation environment. Gaze prediction is a critical component in applications such as foveated rendering, which reduces rendering latency by dynamically allocating computational resources based on user attention. Therefore, our proposed privacy mechanism has been integrated and demonstrated in the context of foveated rendering tasks, similar to prior works [25, 38]. Specifically, we demonstrate DL-based gaze prediction across various masking ratios (e.g., 25%, 50%, and 75%), achieving real-time gaze predictions while simultaneously ensuring user privacy during gameplay. Our proposed privacy mechanism is implemented as a Unity plugin written in (*C#*) and loaded via MelonLoader [45], a universal mod loader for Unity, enabling straightforward integration into Unity-based XR games. For comparison, we evaluate the effectiveness of our method against a traditional DP approach for different privacy levels (e.g., low, medium, high) in the gaze prediction tasks.

Virtual Environment and Apparatus: The virtual environment was developed in Unity 6 with the Universal Render Pipeline (URP) [68], featuring a custom forest scene populated with wildlife (e.g., bear, wolf) for realistic visual context. User interaction was managed via the XR Interaction Toolkit [69], which enabled both teleportation and smooth controller-based locomotion for intuitive scene exploration. Gaze-contingent foveated rendering was implemented using URP Shader Graph [67] with four parameters: *Fovea Center* for gaze alignment, *Fovea Radius* for the sharp inner region, *Edge Softness* for transition smoothness, and *Blur Strength* for peripheral blur intensity. For deploying both our proposed privacy mechanism and the DP-based privacy mechanism, we utilized the HTC Vive Pro Eye headset, which offers a display resolution of 2880 × 1600 pixels per eye and a refresh rate of 90 Hz. The headset's integrated eye-tracking hardware provided binocular gaze data at 120 Hz via the HTC SRanipal SDK in conjunction with the Tobii integration [2], enabling real-time gaze position updates at the rendering pipeline's 90 Hz refresh rate. Rendering computations were executed on a host PC equipped with an Intel Core i9 CPU, 128 GB of memory, and an NVIDIA GeForce RTX 3090 GPU to sustain stable frame rates during trials, with foveation and privacy mechanisms operating transparently within the same rendering pipeline.

GazeProtector Privacy Mechanism, DL-enabled Gaze Prediction Utility Model, and Inference Phase: After training and validation, the CAM-guided Transformer encoder is deployed on the XR headset to perform real-time privacy-preserving encoding. Specifically, the trained encoder identifies utility-relevant regions from the streaming eye-tracking data and applies feature masking on-device before data transmission. The masked representations are then forwarded to a trained Transformer-based utility model deployed and running on a cloud server for real-time gaze prediction. During inference, streaming eye-tracking data are processed using a sliding-window mechanism with fixed-length segments, each of which predicts the gaze position 100 ms into the future. This proposed privacy mechanism ensures that only utility-specific information is transmitted to the cloud, while privacy-sensitive attributes are effectively suppressed without compromising task utility. For the traditional DP-based method, calibrated DP noise is injected into the streaming eye-tracking data under varying privacy budgets (e.g., low ($\varepsilon = 5$), medium ($\varepsilon = 3$), and high ($\varepsilon = 1$)), and the privatized data are subsequently used by the same cloud-deployed utility model for real-time gaze prediction. We employ the Transformer model as both the encoder for region identification and masking and as the utility network for gaze prediction tasks, since it demonstrated superior privacy–utility performance compared to other DL models, as shown in Tables **??**, 4. The deployed Transformer encoder on the HTC Vive Pro Eye headset occupies 30 MB of storage, whereas the cloud-based utility model requires 63 MB. The corresponding training times are 532 minutes for the encoder and 732 minutes for the utility model. Figure 3 shows the real-time gaze prediction for no privacy, our proposed GazeProtector, and the traditional DP-based method in the foveated rendering tasks. In the XR simulation, the proposed GazeProtector mechanism achieves real-time inference with average prediction latencies of 1.89 ms, 1.94 ms, and 1.96 ms for 25%, 50%, and 75% masking ratios, respectively. In contrast, the DP-based method with a Transformer for privacy budgets ϵ low, medium, and high achieves prediction latencies of 3.89 ms, 3.97 ms, and 4.05 ms, respectively, for the same 100 ms prediction horizon. We observe that, overall, the proposed GazeProtector privacy mechanism achieves inference up to $\approx 2\times$ faster than the DP-based approach, owing to its deterministic CAM-guided masking, which eliminates noise-sampling overhead. It ensures real-time operation ($< 2ms$ latency) while maintaining high gaze-prediction accuracy. In contrast, DP-based methods incur higher latency due to stochastic perturbation and signal degradation. These results demonstrate that GazeProtector provides a superior privacy–utility–efficiency trade-off for deployment in XR environments.

8 Discussion

Our experimental results show that the *GazeProtector* effectively protects user privacy in gaze-prediction tasks against strong adversaries while preserving task utility. By using CAM-guided spatial-temporal feature disentanglement, the model identifies task-relevant gaze regions and selectively masks identity-bearing cues. This substantially reduces re-identification with only negligible utility loss compared to the no-privacy mechanism. For instance, at a 75% masking ratio, *GazeProtector* reduces re-identification under the RBFN model by approximately 83.2% on DGaze and 83.4% on OpenEDS2020 datasets. In terms of utility, the proposed privacy mechanism incurs negligible performance loss: with 25% masking, the Transformer attains an MAE of 3.71° on DGaze and 3.81° on OpenEDS2020 datasets, which are almost identical to those obtained with no privacy mechanism, as shown in Table **??**. In addition, *GazeProtector* achieves up to 2× faster inference than SOTA DP-based privacy methods while maintaining DP-comparable resistance under RDA, with higher utility and real-time suitability. These results also highlight that unprotected XR data (e.g., eye-tracking) can act as near-unique user identifiers, creating serious privacy risks under RDA conditions, as shown in Tables **??** and 4.

Prior privacy-preserving methods mainly rely on post-hoc sanitization or statistical noise injection. However, such approaches often obscure task-relevant features, degrade accuracy, increase

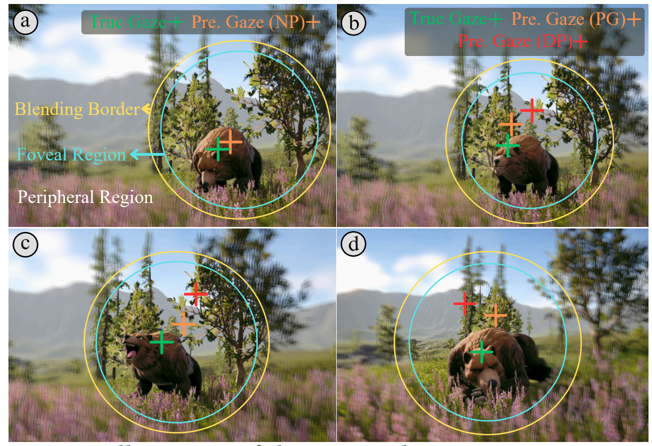

Figure 3: Illustration of the proposed *GazeProtector* privacy mechanism integrated with real-time DL-enabled gaze prediction for foveated rendering tasks. The inner circle denotes the high-fidelity foveal region, the blending border indicates the transition zone, and the outer circle represents the lower-quality peripheral region. The green cross marks the true gaze position; the orange crosses show predictions without privacy (NP) and with *GazeProtector* (PG); and the red cross indicates predictions under the DP-based privacy mechanism. From left to right: (a) baseline without privacy; (b) PG with 25% masking and $\varepsilon = 5$; (c) PG with 50% masking and $\varepsilon = 3$; and (d) PG with 75% PG masking and $\varepsilon = 1$.

latency, and add computational overhead, limiting their suitability for real-time XR applications. In contrast, our approach enforces privacy at the data-acquisition stage by encoding signals during capture and suppressing sensitive attributes before they are exposed to downstream DL models. This enables spatiotemporal separation between utility-relevant features, such as gaze direction, and privacy-sensitive cues, such as user identity, thereby reducing re-identification and attribute inference risks while preserving gaze prediction utility. By preventing raw sensitive data from being digitally exposed, stored, or transmitted to cloud services, the proposed architecture minimizes the attack surface while maintaining the low-latency requirements of real-time XR. Prior studies show that eye-tracking data can reveal sensitive attributes, including identity, gender, and sexual orientation [11, 12, 73]. Consistent with this evidence, our results confirm that raw eye-tracking signals are highly vulnerable to re-identification. *GazeProtector* addresses this risk by suppressing privacy-bearing features during sensor readout, while an anchor loss further promotes task-specific disentanglement by encouraging CAMs to focus on distinct spatio-temporal regions for privacy and utility representations. Across masking ratios, our method maintains low MAE and compares favorably with SOTA privacy-preserving gaze approaches [5, 11, 12, 14, 16, 17, 23, 28, 37, 72, 74, 75]. For instance, on the DGaze dataset,, David et al. [11] reported gaze-prediction errors of 6.5° for k-same-synth (k=6) and 7.3° for kalε-ido (ε=5, r=10, w=50), while Wu et al. [75] reported MAEs of 4.67° for task-oriented and 9.79° for free-viewing scenarios using DP-based privacy. In contrast, with only 25% masking on DGaze, our Transformer model achieves an MAE of 3.71° while reducing re-identification risk and maintaining efficient inference. Overall, these results show that *GazeProtector* provides strong resilience against RDA while preserving high utility, advancing on-device privacy-preserving computation for real-time, user-centric XR.

Generalizability and Applicability of the Proposed GazeProtector Mechanism to Other AI-Driven XR Utility Applications: While our proposed GazeProtector privacy mechanism is designed for DL-based gaze prediction tasks, its on-device encoding framework can be seamlessly extended to other ML/DL-based XR utility applications, such as emotion classification [1, 33], attention prediction [35, 63], activity recognition [33], cybersickness prediction [31, 32], and cognitive load estimation [19, 63], that rely on multimodal XR sensor streams, including hand tracking, head motion, and physiological signals [32, 63]. These data, while essential for user-state modeling, also embed sensitive personal information that can be exploited for re-identification or behavioral profiling. By performing on-device encoding to extract only task-relevant representations and suppress privacy-sensitive attributes before transmission or storage, GazeProtector ensures privacy protection across diverse XR modalities without degrading model accuracy or real-time performance, making it a generalizable and efficient privacy solution for next-generation AI-driven XR systems.

9 Limitations and Future Work

Our proposed framework has a few limitations. First, our identification results are based on an RBFN model; while this choice does not affect the formal privacy guarantees, prior work has explored stronger re-identification models (e.g., Random Forest [61], KNN [37], and MLPs [37, 39]), and evaluating these alternatives is an important direction for future research. However, multi-modal XR sensor data, such as head pose and motion [54], can significantly enhance gaze prediction but also pose substantial privacy risks. Specifically, combining attention data with content can reveal sensitive attributes such as biases, sexual orientation, and emotional states [47]. These sensors provide rich, highly personal, sensitive biometric information. Thus, future work should extend our on-device privacy mechanism to mitigate risks posed by multimodal data in real-world XR deployments. Furthermore, recent studies have leveraged pre-trained large language models, such as the GLAM [49], GPT-4o, and Gemini 2.0 [42] for gaze prediction tasks. Thus, applying our proposed privacy mechanism to these gaze prediction models and analyzing their privacy-utility trade-off would be interesting. While we successfully deployed our privacy-preserving mechanism on a Transformer model for real-time gaze prediction within an XR headset, we were unable to conduct a user study due to time, resource, and budget constraints. We plan to conduct a user study in the future to validate the effectiveness of our proposed privacy mechanism in practical, real-world settings.

10 Conclusion

This work introduced *GazeProtector*, an on-device encoding-based privacy mechanism that preserves utility-relevant gaze features while suppressing identity-bearing and sensitive cues to defend against RDA-based re-identification in XR. Using two open-source XR datasets (e.g., DGaze and OpenEDS2020), we showed that *GazeProtector* maintains strong gaze-prediction utility while protecting privacy. For instance, with 25% masking, the Transformer achieves MAE values of 3.71° on DGaze and 3.81° on OpenEDS2020, remaining close to the no-privacy baselines of 3.45° and 3.68°, respectively. Under the same setting, *GazeProtector* reduces re-identification rates by approximately 63.6% and 64.4% on DGaze and OpenEDS2020 datasets, respectively, demonstrating effective privacy protection. We further deploy the mechanism on an HTC Vive Pro headset to enable real-time, privacy-preserving gaze prediction during gameplay. Experimental results show that the proposed on-device privacy mechanism significantly reduced inference time (up to $\approx 2\times$) compared to the traditional DP-based privacy mechanism while maintaining high utility. Overall, *GazeProtector* advances real-time, privacy-aware, and user-adaptive gaze prediction for consumer XR headsets by effectively balancing privacy, utility, and efficiency.

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